

PRANAW GAUTAM

AI/ML Engineer | Data Scientist | Python, FastAPI & Flutter

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SUMMARY

AI/ML Engineer at Vibe6 Digital building a recommendation system for a client's audio-storytelling app, from raw event data and offline evaluation to an on-device Flutter prototype. Works in Python, pandas, scikit-learn and SQL, and proves results on held-out data before shipping. MCA, GPA 8.65/10.

TECHNICAL SKILLS

Machine Learning: scikit-learn, recommender systems, collaborative filtering, content-based filtering, feature engineering, model evaluation (NDCG, precision/recall)

Data: Python, pandas, NumPy, SQL, EDA, statistical analysis, ETL pipelines, SQLAlchemy, Matplotlib, Seaborn

Apps & APIs: FastAPI, REST APIs, Flutter, Dart, React

BI & Databases: Power BI (DAX), MySQL, PostgreSQL

Tools & Cloud: Git, GitHub Actions, pytest, Docker, AWS (EC2, S3)

EXPERIENCE

AI/ML Engineer | Vibe6 Digital LLP, Indore Aug 2026 – Present

- Built the recommendation pipeline in Python (pandas, scikit-learn) for a client's audio-storytelling app: cleaned a raw Firestore export, engineered user–title features and set up leakage-free, time-based splits.
- Benchmarked 8 approaches (popularity, TF-IDF content-based, item-item collaborative filtering, hybrid); weighted popularity beat heavier models on sparse data (NDCG@10 0.456 vs 0.419), steering the design to simpler ranking.
- Designed a shelf-based home screen that reached 87.4% of held-out users with a title they went on to play, vs 82.6% for a single list of equal length (McNemar $p = 0.0018$).
- Trained a logistic-regression ranker that lifted top-10 hit rate from 77.9% to 80.1% (562 users, $p = 0.02$).
- Caught an 18+ title being recommended to 99% of users who had not confirmed their age; enforced age gating in the ranking and shelf logic, cutting that to zero.
- Built the Flutter prototype that ranks titles on-device from exported model weights and explains every card; parity tests keep the Dart engine identical to the Python reference.

Data Science Intern | Shanti Infosoft LLP Jan 2026 – Jun 2026

- Built reusable preprocessing pipelines in Python (pandas, NumPy) as modular cleaning and transformation components.
- Ran exploratory data analysis to surface patterns and data-quality issues; presented findings with Matplotlib and Seaborn.
- Debugged and optimized pipeline scripts for reliability; took part in code reviews, updates and testing.

PROJECTS

Personalized Audio Recommender | Python, scikit-learn, FastAPI, pytest 2026 · Open source

- Open-source recommender on synthetic data: 7 models behind one interface, an explainable shelf home screen, a FastAPI service and demo, 79 tests with CI; tuned hybrid NDCG@10 0.177 vs 0.160 for popularity.

Space Launch Analytics | Python, MySQL, SQLAlchemy, Power BI 2026

- Python ETL that validates and loads launch, rocket and ISRO data into a 4-table MySQL warehouse with engineered KPIs such as cost per kg and payload utilization.
- 6-page Power BI report with DAX on 7,500 launches since 1957: 83.05% success rate, 35 launch sites, 26 operators, cost trends and an ISRO deep-dive.

Air Quality & Pollution Tracking | Python, pandas, SQL, SQLAlchemy 2026 – Present

- ETL pipeline unifying 141,752 rows of Indian air-quality data from 3 sources, with a per-pollutant data-quality report and SQL analyses of city, AQI-category and seasonal trends; now adding 24–48-hour AQI forecasting.

EDUCATION

Master of Computer Applications (MCA) | AKS University, Satna | GPA 8.65/10 2024 – 2026

Bachelor of Computer Applications (BCA) | APS University, Rewa 2015 – 2018

CERTIFICATIONS & ACHIEVEMENTS

- National Finalist (Top 10), GOER 2025 – Institute of Risk Management (India Affiliate) | RMAT score: 96% (240/250), 2025
- Advanced Program on Amazon Web Services (332 hours, Grade A1) – Anudip Foundation, supported by Accenture, 2025
- Data Science Bootcamp – GeeksforGeeks | Data Science Internship Certificate – Shanti Infosoft LLP, 2026